

Fundamentals of Linear Algebra – Assignment Answers

Question 1: Can you identify whether each of the following is a Scalar, Vector, Matrix, or Tensor?

Answer:

- a) 5 is a Scalar because it has only one value.
- b) [2, 4, 6] is a Vector because it is a one-dimensional array.
- c) [[1, 2], [3, 4]] is a Matrix because it has rows and columns.
- d) A $256 \times 256 \times 3$ colour image is a Tensor because it has three dimensions.

Question 2: What are the dimensions of these matrices?

Answer:

- a) 2×3
- b) 4×1
- c) 3×2

Question 3: Given $A = \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$, find $A + B$, $A - B$, and $2A$.

Answer:

$$A + B = \begin{bmatrix} 6 & 6 \\ 9 & 15 \end{bmatrix}$$

$$A - B = \begin{bmatrix} -2 & 4 \\ -3 & -1 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & 10 \\ 6 & 14 \end{bmatrix}$$

Question 4: What is the transpose of $\begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$?

Answer:

$$\text{Transpose} = \begin{bmatrix} 2 & 8 \\ 4 & 10 \\ 6 & 12 \end{bmatrix}$$

Question 5: What is the transpose of $\begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 4 & 6 \end{bmatrix}$?

Answer:

$$\text{Transpose} = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 5 & 6 \end{bmatrix}$$

Question 6: Can you multiply these matrices?

Answer:

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Question 7: Determine whether the following matrix multiplications are possible. If they are possible, write the dimensions of the resulting matrix.

- a) Possible. Result = 2×4
- b) Not possible.
- c) That is possible. The result is 4×5 .

Question 8:

Given

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$\text{Answer: } AB = \begin{bmatrix} 17 & 20 \\ 41 & 48 \end{bmatrix}$$

Question 9: Verify whether matrix multiplication is commutative.

Compute

AB

and

BA

Answer: $AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$

$BA = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}$

So, matrix multiplication is not commutative.

Question 10:

Answer: Magnitude = 5

Question 11:

Answer: Magnitude = $\sqrt{65} \approx 8.06$

Question 12:

Answer: Dot product = 21

Question 13:

Answer: The vectors are orthogonal because their dot product is 0.

Question 14:

Answer: Normalised vector = [0.6, 0.8]

Question 15:

Answer: Euclidean distance = 10

Question 16:

Answer: The vectors are not linearly independent because one vector is twice the other.

Question 17:

Answer: Yes. The vector belongs to the span of the given vectors.

Question 18:

Answer: A basis for $\mathbb{R}^2$ is $\{[1,0], [0,1]\}$.

Question 19:

Answer: Rank = 1

Question 20:

Answer: Rank = 2

Question 21:

- a) Orthogonal
- b) Orthogonal
- c) Not orthogonal

Question 22:

Answer: Cosine similarity = 0.98. This means the two vectors point in almost the same direction.

Question 23:

- Orthogonal vectors: Vectors whose dot product is zero.
- Orthonormal vectors: Orthogonal vectors with unit length.
- Basis: A set of linearly independent vectors that span a vector space.
- Vector space: A collection of vectors that is closed under vector addition and scalar multiplication.

